3D Vectors & the Cross Product

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

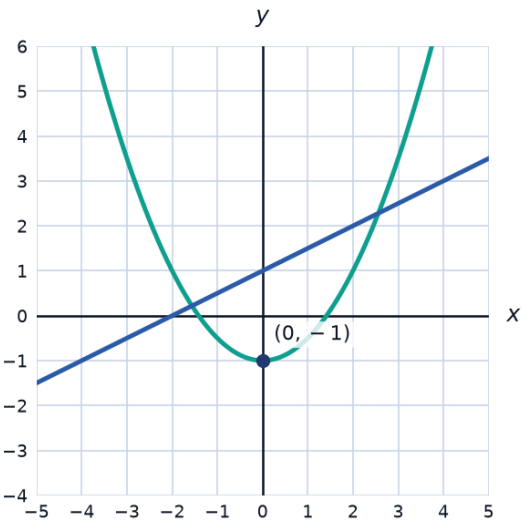
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Dot Product and Orthogonality**

Consider the tetrahedron OABC with vertices O(0, 0, 0), A(2, 1, 3), B(1, 2, -1), and C(k, k+1, k-1). Given that the volume of the tetrahedron is 7 cubic units, find the possible values of k.

2 **[Foundational] • Target: The Vector Cross Product Formula**

Consider the points A(2, 3) and B(8, 7) in the Cartesian plane. The vector $\vec{AB}$ can be represented as **v**. Find the components of **v**. .



3 **[Foundational] • Target: Area of Triangles and Parallelograms**

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

4 **[Foundational] • Target: Scalar Triple Product & Volume**

Consider the points $A(1, 2)$ and $B(4, 6)$. Find the vector $\vec{AB}$.

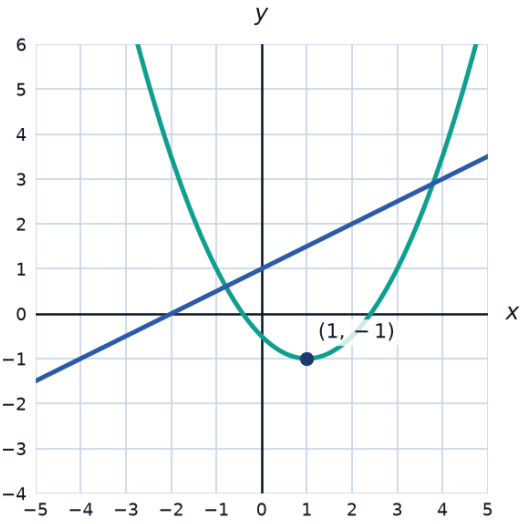
5 **[Foundational] • Target: Dot Product and Orthogonality**

Consider the points $P(2, -1, 3)$ and $Q(5, 3, 0)$. Find the vector $\vec{PQ}$.

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6 [Foundational] • Target: The Vector Cross Product Formula

Consider the points $A = (1, 2, 3)$ and $B = (4, 0, -1)$. Find the vector $\vec{AB}$.



7 [Foundational] • Target: Area of Triangles and Parallelograms

A line has vector equation $\mathbf{r} = (-2;\ -2; 0) + t(-2;\ -2;\ -4)$. Find the point when $t = 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Scalar Triple Product & Volume

A line has vector equation $\mathbf{r} = (-1; 4; 6) + t(-1; 4; 2)$. Find the point when $t = -4$.

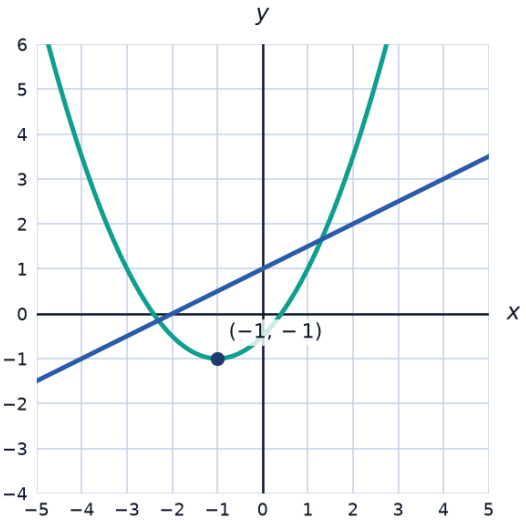
9 [Intermediate] • Target: Dot Product and Orthogonality

A line has vector equation $\mathbf{r} = (0;\ -3; 5) + t(3;\ -4;\ -3)$. Find the point when $t = 3$.

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10 [Intermediate] • Target: The Vector Cross Product Formula

A line has vector equation $\mathbf{r} = (6;\ -6;\ -6) + t(1;\ -1;\ -2)$. Find the point when $t = 0$.



11 [Intermediate] • Target: Area of Triangles and Parallelograms

A line has vector equation $\mathbf{r} = (4;\ -4;\ -6) + t(2;\ -3;\ -3)$. Find the point when $t = -2$.

12 [Intermediate] • Target: Scalar Triple Product & Volume

A line has vector equation $\mathbf{r} = (-2;\ 1;\ -2) + t(-1;\ 4;\ 4)$. Find the point when $t = 1$.

13 [Intermediate] • Target: Dot Product and Orthogonality

A line has vector equation $\mathbf{r} = (4;\ 3;\ 3) + t(2;\ 2;\ -1)$. Find the point when $t = 3$.

14 [Intermediate] • Target: The Vector Cross Product Formula

Use the compound angle formula for cosine to show that $\cos(180^\circ - x) = -\cos x,$ then find $\cos 150^\circ$. .

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

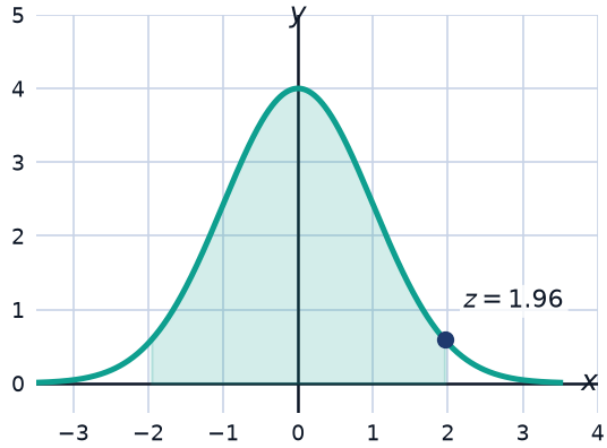
15 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 1.04 Partial Fractions

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-5;\ -2;\ 2)$ passing through $P(-3,\ -4,\ -3)$.

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16 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-6;\ -6;\ -3)$ passing through $P(2, 6, -2)$.



17 [Challenging] • Target: Dot Product and Orthogonality • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-1; 3; 4)$ passing through $P(-5, -6, -1)$.

18 [Challenging] • Target: The Vector Cross Product Formula • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (1;\ -3; 4)$ passing through $P(-6, 6, -1)$.

19 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-4;\ -6; 4)$ passing through $P(0, 4, -2)$.

20 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-3; 5;\ -5)$ passing through $P(-5, 5, -3)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Dot Product and Orthogonality

Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [2, -3, 5]$ and $\mathbf{v} = [4, 1, -2]$.

a. -5

b. 5

c. 15

d. -1

22 [Foundational] • Target: Dot Product and Orthogonality

For what value of k are the vectors $\mathbf{a} = [k, 3, -2]$ and $\mathbf{b} = [4, -2, 5]$ orthogonal (perpendicular)?

a. 4

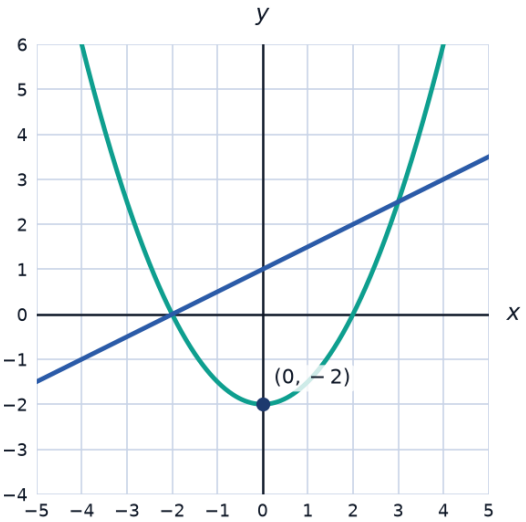
b. 2

c. -4

d. 5

23 [Foundational] • Target: Dot Product and Orthogonality

Find the angle θ between $\mathbf{u} = [1, 1, 0]$ and $\mathbf{v} = [0, 1, 1]$.



a. $\frac{\pi}{3}$ (60°)

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{6}$ (30°)

d. $\frac{\pi}{2}$ (90°)

24 [Foundational] • Target: The Vector Cross Product Formula

Compute the cross product $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.

a. $[-3, 6, -3]$

b. $[3, -6, 3]$

c. $[-3, -6, -3]$

d. $[7, -2, -3]$

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25 [Foundational] • Target: The Vector Cross Product Formula

What is the geometric relationship between the cross product vector $\mathbf{u} \times \mathbf{v}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$?

a. $\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$

b. $\mathbf{u} \times \mathbf{v}$ is parallel to $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} \times \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$

d. $\mathbf{u} \times \mathbf{v}$ lies in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$

26 [Foundational] • Target: The Vector Cross Product Formula

If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, what does this indicate?

a. The vectors are parallel (collinear)

b. The vectors are perpendicular

c. The vectors have unit magnitude

d. The vectors are identical

27 [Foundational] • Target: Area of Triangles and Parallelograms

Find the area of the parallelogram spanned by vectors $\mathbf{a} = [2, 0, 1]$ and $\mathbf{b} = [1, 3, -2]$.

a. $\sqrt{59}$

b. $\sqrt{35}$

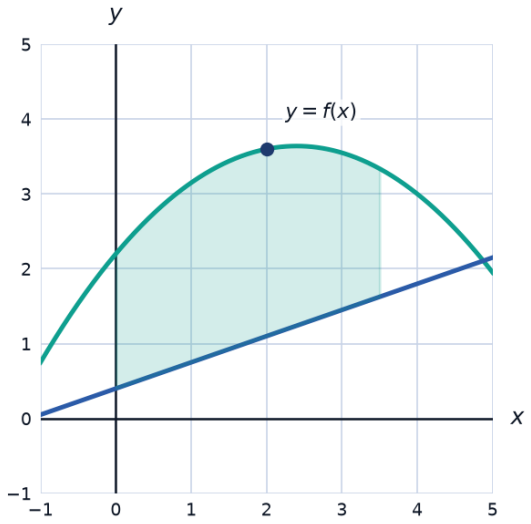
c. 14

d. $\frac{\sqrt{59}}{2}$

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28 [Foundational] • Target: Area of Triangles and Parallelograms

Find the area of the triangle with vertices $A(1, 0, 0), B(0, 2, 0), C(0, 0, 3)$.



a. $\frac{7}{2}$

b. 7

c. $\sqrt{14}$

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Area of Triangles and Parallelograms

If two vectors have lengths $|\mathbf{u}| = 6$ and $|\mathbf{v}| = 5$, and the angle between them is 30°, what is $|\mathbf{u} \times \mathbf{v}|$?

a. 15

b. $15\sqrt{3}$

c. 30

d. $30\sqrt{3}$

30 [Intermediate] • Target: Scalar Triple Product & Volume

What geometric quantity is given by the absolute value of the scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$?

a. The volume of the parallelepiped formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

b. The surface area of the prism formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

c. The length of the main diagonal vector

d. The perimeter of the base

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31 [Intermediate] • Target: Scalar Triple Product & Volume

Calculate the volume of the parallelepiped with edges $\mathbf{a} = [1, 0, 0]$, $\mathbf{b} = [0, 2, 0]$, $\mathbf{c} = [1, 1, 3]$.

a. 6

b. 3

c. 12

d. 2

32 [Intermediate] • Target: Scalar Triple Product & Volume

If $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, what does this reveal about the three vectors?

a. The three vectors are coplanar (lie in the same plane)

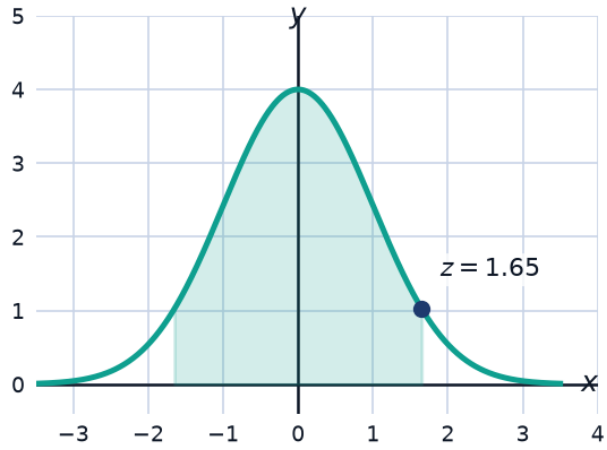
b. The three vectors are mutually perpendicular

c. At least one vector must be the zero vector

d. The vectors form an equilateral triangle

33 [Intermediate] • Target: Dot Product and Orthogonality

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (5; -5; 5)$ passing through $P(-6, 2, 2)$.



a. $5x - 5y + 5z = -28$

b. $5x - 5y + 5z = -27$

c. $5x - 5y + 5z = -29$

d. $5x - 5y + 5z = -30$

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34 [Intermediate] • Target: The Vector Cross Product Formula

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (6; 5; -2)$ passing through $P(-1, -1, -6)$.

a. $6x + 5y - 2z = -2$

b. $6x + 5y - 2z = 0$

c. $6x + 5y - 2z = -1$

d. $6x + 5y - 2z = 1$

35 [Intermediate] • Target: Area of Triangles and Parallelograms

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (4; -2; 6)$ passing through $P(-3, 6, 1)$.

a. $4x - 2y + 6z = -18$

b. $4x - 2y + 6z = -21$

c. $4x - 2y + 6z = -19$

d. $4x - 2y + 6z = -15$

36 [Intermediate] • Target: Scalar Triple Product & Volume

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (3; -6; -6)$ passing through $P(1, 0, -1)$.

a. $3x - 6y - 6z = 11$

b. $3x - 6y - 6z = 9$

c. $3x - 6y - 6z = 6$

d. $3x - 6y - 6z = 8$

37 [Intermediate] • Target: Dot Product and Orthogonality

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (3; 2; -1)$ passing through $P(6, 2, -5)$.

a. $3x + 2y - z = 25$

b. $3x + 2y - z = 28$

c. $3x + 2y - z = 27$

d. $3x + 2y - z = 30$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Vector Cross Product Formula • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Using the compound angle formula, expand $\sin(x + 90^\circ)$.

a. $\cos x$

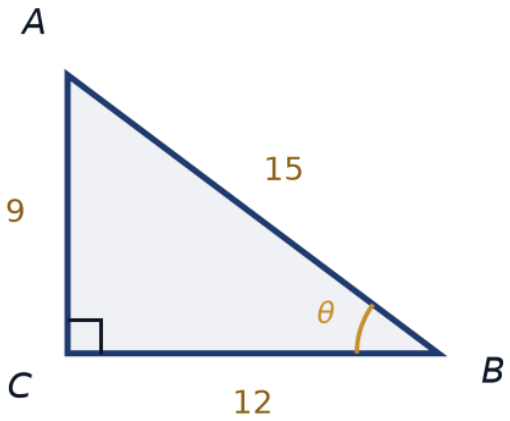
b. $-\cos x$

c. $\sin x$

d. $-\sin x$

39 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

The diagram shows two similar triangles ABC and ADE, where D lies on AB and E lies on AC. It is given that AD = 6 cm, DB = 4 cm, and DE = 12 cm. Find the length of BC.



a. 18 cm

b. 20 cm

c. 16 cm

d. 24 cm

40 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the points $A(1, -2, 3)$, $B(4, 0, 5)$, and $C(p, 1, 2)$. If the vectors $\vec{AB}$ and $\vec{AC}$ are perpendicular, find the value of p.

a. $p = -5$

b. $p = -4$

c. $p = 4$

d. $p = 5$

41 [Challenging] • Target: Dot Product and Orthogonality • Cross-Lesson: 4.02 The Poisson Distribution

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, 3, -1)$, $B(-2, 0, 3)$, and $C(-4, 3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 3

b. 2

c. 8

d. 7

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42 [Challenging] • Target: The Vector Cross Product Formula • Cross-Lesson: 5.01 Further Integration Techniques

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, 1, -4)$, $B(1, 4, -2)$, and $C(-1, -3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 6

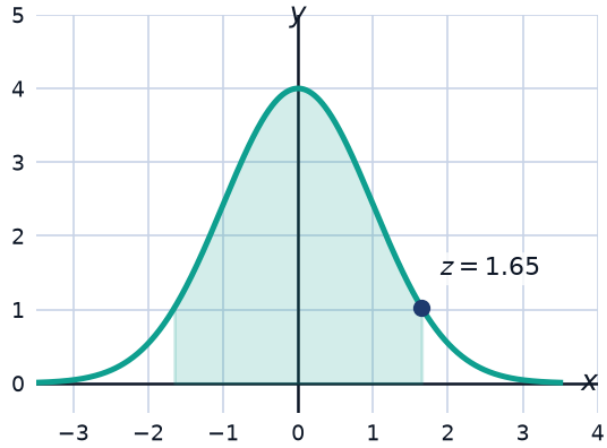
b. 9

c. 7

d. 14

43 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 5.02 Differential Equations

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, -3, 1)$, $B(-3, 4, 2)$, and $C(-4, -2, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.



a. -7

b. -17

c. -10

d. -13

44 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, -1, -4)$, $B(-4, 4, 4)$, and $C(1, 2, 0)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 51

b. 39

c. 52

d. 37

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45 [Challenging] • Target: Dot Product and Orthogonality • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, -3, 4)$, $B(2, -1, 2)$, and $C(-3, 2, 1)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 4

b. 24

c. 20

d. 18

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